High-Value Customer Identification & Social Influence Mapping

An automated algorithmic workflow for resolving KYC identities against public social graphs, quantifying asymmetric reputational risk, and orchestrating priority service routing across a 50-lakh customer base.

CLIENT ORGANIZATION
Axis Securities Limited (AxisDirect)

DELIVERY PARTNER
HashCase (HashByte Pvt Ltd)

DOCUMENT VERSION & STATUS
Version 2.4 — Production Baseline

ENGAGEMENT PERIOD
Q3–Q4 2024 (Pilot Run & Production Deployment)

TARGET DEMOGRAPHIC
50,00,000 KYC Records (Retail, HNI, Trading)

REGULATORY & INFOSEC REVIEW
Compliant with DPDP Act 2023 & SEBI KYC Norms

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1. Executive Summary & Strategic Rationale

AxisDirect maintains Know Your Customer (KYC) records for approximately **50 lakh (5 million) customers** across retail equity, derivatives, currency, and mutual fund advisory accounts. Within this base exists a critical micro-segment whose public reach and social credibility make their individual service experience consequential far beyond the commercial value of their trading book.

If a venture-backed tech founder with 75,000 LinkedIn followers, a prominent finance creator with 2,50,000 YouTube subscribers, or a financial daily editor experiences a failed order execution, an unresolved margin penalty, or a delayed KYC unlock and subsequently posts a screenshot publicly, the resulting brand exposure is asymmetric, immediate, and nearly impossible to contain once viral momentum takes hold.

HashCase (HashByte Pvt Ltd) was engaged to build and deploy an automated data processing workflow that systematically reconciles AxisDirect's KYC database against public professional and social graphs. The resulting intelligence layer assigns an algorithmic **Composite Influence Score (CIS)** and categorizes high-exposure individuals into distinct operational service tiers.

The output supports service prioritisation and reputational risk management. It functions as an operational protective guardrail rather than a marketing or sales solicitation tool.

2. The Multi-Dimensional Scoring Framework

To quantify reputational exposure objectively, each matched profile is evaluated across four core attributes. Rather than treating raw follower counts as an isolated metric, the model contextualizes reach against institutional credibility and topical relevance to capital markets.

ATTRIBUTE	DEFINITION & RATIONALE	PRIMARY DATA SIGNALS EXTRACTED	MATHEMATICAL TREATMENT
Reach (R)	The absolute scale of the addressable audience the customer can broadcast to directly.	Followers on LinkedIn, X (Twitter), Instagram; subscribers on YouTube channels.	Logarithmically compressed: $\log_{10}(R+1)$. Prevents extreme outlier distortion.
Authority (A)	The institutional credibility and weight attributed to the individual's public commentary.	Platform verification badges, MCA-21 Director filings, C-suite/Partner titles, media citations.	Standardized index [0–100] based on credential hierarchy and third-party verification.
Relevance (M)	Topical overlap between the customer's audience and retail broking/wealth markets.	Bio keyword analysis, content topic modeling (taxation, trading, equity research, macroeconomics).	Multiplicative modifier [0.50x to 1.35x]. Heavily discounts non-business/lifestyle accounts.
Grievance Vulnerability	Real-time responsiveness to service friction and historical propensity for public escalation.	Integration with active social listening webhooks (Sprinklr / X Enterprise API) detecting brand mentions.	Dynamic event-based CRM queue override rather than a static additive penalty.

3. Key Customer Archetypes in the AxisDirect Base

Analysis of the confirmed high-influence cohort revealed five recurring archetypes within AxisDirect's demographic base. Each archetype presents distinct escalation patterns and demands a calibrated operational response.

CUSTOMER ARCHETYPE	DEMOGRAPHIC & IDENTITY INDICATORS	PRIMARY ESCALATION CHANNELS	REPUTATIONAL RISK VECTOR
1. Startup Founders & Tech CXOs	Aged 28–48; corporate email domains; Bengaluru, Mumbai, NCR, Pune; LinkedIn network > 15,000; MCA active directorships.	LinkedIn posts, X quote-tweets tagging executive leadership.	High peer-network resonance. Tech and startup ecosystems quickly mobilize and share grievances.
2. Finfluencers & Trading Educators	Active YouTube/Instagram creators; bio keywords: *Option Trader, Nifty Analyst, Wealth Coach*; 50k–1M+ followers.	Live YouTube streams, Instagram stories, screenshots on X / Telegram channels.	Immediate mass-retail audience. Platform downtime or order latency screenshots directly deter prospective accounts.
3. Financial Journalists & Media Editors	Reporters, columnists, and anchor staff at major publications (Mint, ET, Moneycontrol, CNBC-TV18, Bloomberg).	X threads, direct editorial mentions, industry WhatsApp/Signal groups.	Direct access to mainstream media amplification. A service bug can become an investigative article on broking compliance.
4. Senior Legal & Regulatory Figures	Senior Advocates, NCLT/High Court counsel, retired administrative/regulatory officials, Big-4 Tax Partners.	Formal legal notices, direct letters to SEBI / Banking Ombudsman, high-level board escalations.	Systemic compliance and regulatory audits rather than viral public social outrage.
5. Derivatives Community Leaders	High-frequency option writers running popular Discord servers or private trading groups (1,000–10,000 active traders).	Community broadcasts, collective support ticket spamming, viral X campaigns.	Multiplied operational pressure. An outage experienced by one leader triggers simultaneous panic across hundreds of clients.

4. Technical Funnel Architecture & Methodology

Querying live public social graphs across a large customer base is cost-prohibitive and introduces massive false positives. The HashCase architecture functions as a precision-engineered funnel designed around two immutable constraints:

- **The Indian Name Collision Paradox:** Common names appear thousands of times within the customer base. Name alone is zero proof of identity; matches require multi-attribute corroboration.
- **Temporal Decay of KYC Declarations:** KYC data captures an individual at onboarding. Historical fields serve as anchor seeds, but current authority is resolved from live public records.

STAGE 1	**Data Ingestion, Transliteration Normalisation & Salted Hashing:** Name standardisation using Indian phonetic models (variants like *Rajesh/Rajeshh, Dikshit/Dixit, Banerjee/Bandopadhyay*). Parsing addresses into standardized PIN, City, and State. Validating corporate domains against free webmail. Salted SHA-256 pseudonymisation creates a decoupled processing environment.
STAGE 2	**High-Recall Candidate Shortlisting:** Low-cost heuristic filtering using native KYC fields: (1) Occupation regex across 250+ senior roles (Director, Founder, Partner, MD, Editor, Advocate); (2) Verified corporate email domains; (3) Declared annual income or Demat account vintage; (4) Tier-1 metropolitan concentration; (5) Existing Burgundy/HNI relationship flags.
STAGE 3	**Multi-Attribute Social Graph Discovery:** Automated queries across authorized enterprise endpoints (LinkedIn API, X Enterprise API, Google Knowledge Graph API, MCA-21 Company Registry, Crunchbase). Queries combine strict composite tokens: `[Full Name] + [Declared Employer]`, `[Full Name] + [Corporate Domain]`, and `[Email Handle] + [Location]`.
STAGE 4	**Disambiguation & Deterministic Identity Resolution:** Each candidate match is evaluated against a weighted corroboration matrix. Matches are assigned a Confidence Score between 0.00 and 1.00. Ambiguous cases are routed to a Human-in-the-Loop (HITL) audit queue or pruned entirely.
STAGE 5	**Influence Enrichment, Algorithmic Scoring & CRM Ingestion:** Enrichment of confirmed profiles with follower counts, median post engagement, verification status, and topical tags. Generation of the final Composite Influence Score (CIS) and assignment into operational service tiers.

4.4 The Disambiguation Matrix & Confidence Thresholding

The resolution engine calculates match confidence (C_i) as a weighted sum of normalized corroboration vectors:

CORROBORATION VECTOR	WEIGHT	VALIDATION LOGIC & DISAMBIGUATION RULES
Employer / Org Consistency	**35%**	Fuzzy matching (Levenshtein distance > 0.88) between KYC employer field and public LinkedIn experience graph / MCA directorships.
Username / Email Handle Overlap	**20%**	Jaro-Winkler string similarity between the customer's email username (e.g., *vikram.mehta*) and public social handles (@vikrammehta).
Geographic / City Alignment	**20%**	Exact match of current city or metro cluster between KYC address and public profile location.
Chronological Consistency	**15%**	Validation that declared date of birth aligns plausibly with university graduation years or corporate career milestones.
Name Rarity Weight (IDF)	**10%**	Inverse Document Frequency across the customer base. Unique names require fewer corroborating signals than common names.

Confidence Thresholds & Human-in-the-Loop (HITL) Protocol

- **Automated Acceptance ($C_i \geq 0.85$):** High-confidence deterministic matches. Ingested directly into the influence scoring engine without human intervention.
- **Human-in-the-Loop Review Queue ($0.70 \leq C_i < 0.85$):** Ambiguous profiles where public reach is substantial (>25k followers). Routed to compliance analysts for a brief visual verification; ambiguous remainders purged.
- **Automated Pruning ($C_i < 0.70$):** Discarded immediately. An erroneous false-positive identification is deemed far more damaging than an omitted match.

4.5 Mathematical Formulation of the Composite Influence Score (CIS)

The enriched attributes are normalized and synthesized into a single scalar score CIS ∈ [0, 100]:

```
CIS = MIN(100, [ w_reach · S_reach + w_auth · S_auth + w_eng · S_eng ] × M_relevance)

Where:
• S_reach     = (log10(Total_Audience + 1) / log10(1,000,000)) × 100     [w_reach = 0.40]
• S_auth      = Normalized credential score (Verification, MCA Directorship, CXO)   [w_auth = 0.35]
• S_eng       = Normalized 30-day median interaction rate (replies/shares)          [w_eng = 0.25]
• M_relevance = Industry topical multiplier: 1.35x (Finance/Trading), 1.00x (Business/Tech), 0.50x
(General/Lifestyle)
```

Key Mathematical Characteristics: (1) *Logarithmic Reach Compression:* The marginal difference between 1,000 and 20,000 followers is significant, while the difference between 700,000 and 720,000 is negligible. (2) *Authority Weighting:* High institutional authority outscores lifestyle accounts with passive followings. (3) *Multiplicative Relevance:* Irrelevant audiences are discounted by up to 50%, ensuring capital markets exposure remains the primary sorting criteria.

5. Operational Tiering Framework & Service SLAs

Continuous decimal scores cannot be directly executed by frontline customer service desks. The pipeline categorizes scored customers into distinct operational tiers embedded directly into AxisDirect's CRM (Salesforce / LeadSquared) and ticketing systems.

TIER	REPRESENTATIVE COHORT	OPERATIONAL ESCALATION PATH & CRM PROTOCOL	RESOLUTION SLA STANDARDS
TIER 1 **Executive VIP**	Listed CXOs, prominent finfluencers (>100k relevant reach), senior national business editors, Unicorn founders.	Direct routing to Senior Executive Concierge Desk; dedicated Senior RM; automatic duplicate alert dispatched to Head of Customer Experience (CX); mandatory VP review prior to ticket closure.	**First Response:** < 15 mins **Resolution Target:** < 4 hours **Real-Time Monitoring:** 24/7 active social mention alert.
TIER 2 **High Priority**	Series A/B founders, corporate VPs/Directors, certified research analysts, active trading coaches (15k–100k reach).	Prioritized queue routing in ticketing system; auto-assignment to Level-2 Senior Support Engineers; daily supervisor reconciliation of all open tickets.	**First Response:** < 1 hour **Resolution Target:** < 12 hours **Escalation Threshold:** Alert at 6h unresolved.
TIER 3 **Monitored**	Boutique financial advisors, senior chartered accountants, tech leads, mid-tier commentators (3k–15k reach).	CRM record tagged with dynamic awareness badge; standard routing queue with auto-elevation to Tier 2 if a ticket is reopened or escalated twice.	**First Response:** < 4 hours **Resolution Target:** Standard 24h TAT **Re-open Trigger:** Immediate L2 routing.
STANDARD **General Retail**	Standard retail investors, passive demat holders, non-public accounts.	Standard automated IVR, conversational AI chatbot triage, standard round-robin support queue.	Standard institutional SLAs (24–48 hours turnaround time).

Operational Governance Matrix

To ensure seamless handoffs between automated classification and human execution, the following operational safeguards are enacted:

- **Supervisor Override:** Relationship managers may manually appeal a customer's tier assignment through a one-click CRM justification form.
- **Silent Incident Management:** If a platform outage occurs during market hours (9:15 AM - 3:30 PM), Tier 1 and Tier 2 customers receive proactive status updates before public inquiries materialize.
- **Closed-Loop Audit:** All escalated interactions undergo weekly sample reviews by the CX quality assurance desk to verify SLA adherence and escalation efficacy.